EXPERIMENT NO. 17:

Implement Mobile Price Prediction using appropriate machine learning algorithm

Aim:

To implement a Mobile Price Prediction Model in Python using Random Forest Regression and predict the price of a mobile phone based on its features.

Procedure:

1. **Import the required Python libraries.**
2. **Create a sample mobile price dataset.**
3. **Separate the input features and target price.**
4. **Split the dataset into training and testing sets.**
5. **Create and train the Random Forest Regression model.**
6. **Predict mobile prices using the test data.**
7. **Calculate the R^2 score to evaluate the model.**
8. **Predict the price of a new mobile phone.**

Program:

```
# Mobile Price Prediction using Random Forest Regression

import pandas as pd

from sklearn.model_selection import train_test_split

from sklearn.ensemble import RandomForestRegressor

from sklearn.metrics import r2_score

# Sample Mobile Dataset

data = {

    'RAM': [2, 3, 4, 4, 6, 6, 8, 8, 12, 12],

    'Storage': [32, 32, 64, 128, 128, 256, 128, 256, 256, 512],

    'Camera': [12, 13, 16, 20, 24, 32, 48, 50, 64, 108],

    'Battery': [3000, 3500, 4000, 4500, 4500, 5000, 5000, 5500, 5000, 6000],
```

```
    'Price': [8000, 10000, 13000, 16000, 20000, 25000, 30000, 35000, 45000, 60000]

}

df = pd.DataFrame(data)

# Input features and target

X = df[['RAM', 'Storage', 'Camera', 'Battery']]

y = df['Price']

# Split dataset

X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.2, random_state=42)

# Create Random Forest Regression model

model = RandomForestRegressor(n_estimators=100, random_state=42)

# Train the model

model.fit(X_train, y_train)

# Predict test data

y_pred = model.predict(X_test)

# Calculate R2 Score

score = r2_score(y_test, y_pred)

print("Actual Prices:")

print(y_test.values)

print("\nPredicted Prices:")

print(y_pred.round(2))

print("\nR2 Score:")

print(round(score, 2))

# Predict price of a new mobile

new_mobile = [[8, 256, 50, 5000]]

prediction = model.predict(new_mobile)

print("\nPredicted Price of New Mobile:")
```

print(round(prediction[0], 2))

Output:

```
••• Actual Prices:
    [45000 10000]

    Predicted Prices:
    [36550. 10010.]

    R2 Score:
    0.88

    Predicted Price of New Mobile:
    30700.0
```

Result:

The Mobile Price Prediction Model was successfully implemented in Python using Random Forest Regression. The model predicted mobile prices based on RAM, storage, camera, and battery capacity and successfully estimated the price of a new mobile phone.